DSP · L06

DFT and circular convolution

Cheat sheet · Day 09 · Sun 1 Nov · pack D5

Reference-sheet lines

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DFT (L06)
X(k) = sum_{n=0}^{N-1} x(n) W_N^{kn},   x(n) = (1/N) sum_k X(k) W_N^{-kn},  W_N = e^-
W_4 = [1 1 1 1; 1 -j -1 j; 1 -1 1 -1; 1 j -1 -j];   W_N^{-1} = (1/N) W_N^*
Checks: X(0)=sum x(n);  x(0)=(1/N) sum X(k);  real x => X(N-k)=X*(k)
Freq sampling: N samples of X(w) <-> x_p(n)=sum_m x(n-mN). Aliasing iff N < L.
   One period: write x in rows of N, add columns.
X(k) = X(w) at w=2pi k/N  (if L <= N).   DFT vs DFS: X(k) = N c_k
FFT: direct N^2 mults; radix-2 (N/2)log2 N mults (O(N log2 N))
Properties (mod N):  x((n-l)) <-> X(k)e^{-j2pi kl/N};  x((-n)) <-> X((-k)) = X(N-k)
   x(n)e^{j2pi ln/N} <-> X((k-l));  x* <-> X*((-k));  x1 (*)N x2 <-> X1 X2
   x1 x2 <-> (1/N) X1 (*)N X2;  Parseval: sum|x|^2 = (1/N) sum|X|^2
Circular reversal of {x0,x1,x2,x3} = {x0,x3,x2,x1}
Circular conv: y(n)=sum_m x1(m) x2((n-m))_N. Circulant: col 0 = x2, shift down + wrap
   Shortcut: linear conv, then fold samples n>=N back by N and add.
Linear conv via DFT: zero-pad to N >= L+M-1, Y=XH, IDFT.
   If N too small: first L+M-1-N samples are corrupted.
Deconvolution: X(k)=Y(k)/H(k) - fails where H(k)=0.
Check any convolution: sum y = (sum x1)(sum x2).
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